Number Grammar3: People's understanding of the structure of number words (#255962)

Author(s)
This pre-registration is currently anonymous to enable blind peer-review.
It has 2 authors.

Pre-registered on:
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1) Have any data been collected for this study already?
No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?
This study is part of a larger project on people's understanding of the structure of number words. Specifically, we ask whether the number words that participants generate in a novel number language system conform to phrase-structure rules and a syntactic constraint known as the packing strategy, both of which Hurford (2007, 2011) argued to be universal. We also examine whether the presence of irregularities affects the kinds of structures participants are more likely to produce.

3) Describe the key dependent variable(s) specifying how they will be measured.
In this study, participants from different language backgrounds are asked to produce number words for a set of target numerals in a novel number language system. They learn that this system has only two digit words (representing the numerals 1 and 2) and two power words (representing 3 and 9). After being introduced to the system, participants complete comprehension checks to ensure they understand both categories of number vocabulary and the numerals they represent.

Participants then complete several practice trials, with the amount of training varying depending on the condition they are assigned to, before moving on to the main task. In the actual trials, they complete 11 test items in which they are asked to produce number words in the novel number language system for numerals of varying magnitudes. The target numerals for the actual trials are 12, 15, 21, 24, 27, 28, 29, 36, 54, 55, and 56.

Responses will be coded into three categories:
1. Correct on Syntax + packing: conforming to both Hurford's phrase-structure rules and the Packing Strategy
2. Correct on Syntax-only: conforming to phrase-structure rules but not the Packing Strategy (Packing error)
3. No-syntax: not conforming to phrase-structure rules (Syntax error)

The key dependent variables are:
• Syntax + packing (accuracy): coded as correct or incorrect.
• Error type: categorized as either a packing error or a syntax error.

4) How many and which conditions will participants be assigned to?
We will include four conditions that differ only in the practice trials, following a 2 × 2 design (training with small numbers, Yes/No; training with large numbers, Yes/No). This yields four training types: no training, training on small numbers, training on large numbers, and training on both small and large numbers. The small-number set consists of 4, 6, 7, and 11, while the large-number set consists of 26, 33, 38, and 57.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.
To examine whether participants adhere to the syntax and packing principles, we will first compare their accuracy in conforming to both principles against chance levels using bootstrapped confidence intervals. We will then model syntax+packing accuracy using a mixed-effects logistic regression, including language contrast codes, condition contrast codes, and their interactions as fixed effects, with random intercepts for participants and items. Next, we will extend the model by adding practice-trial performance as a moderator. If any model fails to converge, we will run separate models instead.

To analyze the types of errors participants make and how these vary by language background, we will fit similar mixed-effects logistic regressions but with error type as the dependent variable.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.
• Participants who fail any of the catch trials will be excluded. Two catch trials appear among the test trials. In the catch trials, participants are asked to write the numerical expression in the novel artificial number language for numerals 2 and 9. They would pass the catch trials only if they wrote the correct expressions for 2 and 9.
• Participants who take less than 10 minutes to complete the survey will be excluded.
• Participants whose first and primary language is not one of the target languages will be excluded.
• Participants who did not first learn about numbers (counting, arithmetic) in their primary language will be excluded.

Participants who fail any of these exclusion criteria will be excluded from the analysis.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We plan to recruit up to 20 participants per condition for each language group: English, Chinese, and Turkish.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

We will examine whether participants consistently apply the same syntax across all numerals.
We will also explore whether numeral magnitude influences participants' responses, for example, whether larger numbers lead to syntax shifts.
We will analyze responses that do not fit the "syntax+packing" pattern to see if any intuitive syntax trends emerge.
If the exclusion rate exceeds 30%, we will conduct analyses both with and without the excluded participants.
The design is not specifically intended to examine reaction time, but we will explore the effects of the key variables on reaction time as an additional analysis.